Project 1

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```
knitr::opts_chunk$set(echo = TRUE)

# --- Project 1: Chess Tournament Data Transformation ---

# Author: Radzhana Rabdanova and Chingiz Randanov
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✓ dplyr     1.1.4     ✓ readr     2.1.6
✓ forcats   1.0.1     ✓ stringr   1.6.0
✓ ggplot2   4.0.2     ✓ tibble    3.3.1
✓ lubridate 1.9.5     ✓ tidyr     1.3.2
✓ purrr     1.2.1
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✗ dplyr::filter() masks stats::filter()
✗ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
errors
```

```
library(stringr)
library(DT) # For interactive tables
library(ggplot2)

# 1. Loading
# Update the path to your local file location
path <- "C:/Users/radzh/OneDrive/Desktop/data 607/project1/tournamentinfo.txt"
raw_data <- readLines(path)
```

```
Warning in readLines(path): incomplete final line found on
'C:/Users/radzh/OneDrive/Desktop/data 607/project1/tournamentinfo.txt'
```

```
# 2. Cleaning: removing lines that contain no letters or numbers
clean_data <- raw_data[str_detect(raw_data, "[A-Za-z0-9]")]

# 3. Finding where the first player starts
start_index <- grep("^\\s*1\\s*\\|", clean_data)[1]
if(is.na(start_index)) start_index <- 3

# Skip headers
clean_data <- clean_data[start_index:length(clean_data)]

player_list <- list()

# 4. Parsing loop
for (i in seq(1, length(clean_data), by = 2)) {
  if ((i + 1) > length(clean_data)) break
```

```
  line1 <- str_split(clean_data[i], "\\|")[[1]]
  line2 <- str_split(clean_data[i+1], "\\|")[[1]]

  if (length(line1) < 3) next

  id <- as.numeric(str_trim(line1[1]))
  if (is.na(id)) next

  # Extracting fields
  name <- str_trim(line1[2])
  state <- str_trim(line2[1])
  total_pts <- as.numeric(str_trim(line1[3]))

  # Finding Pre-Rating (numbers after 'R:')
  pre_rating <- as.numeric(str_extract(line2[2], "(?<=R: )\\s*\\d{3,4}"))

  # Finding Opponent IDs
  opp_id_raw <- str_extract_all(clean_data[i], "\\|\\s*[WLD]\\s+(\\d+)")
  opponents <- as.numeric(str_extract(opp_id_raw[[1]], "\\d+"))

  player_list[[id]] <- list(
    Name = name,
    State = state,
    TotalPoints = total_pts,
    PreRating = pre_rating,
    Opponents = opponents
  )
}

# 5. Building the table and calculating the average
final_results <- map_df(player_list, function(p) {
  if (is.null(p)) return(NULL)

  # Getting opponent ratings
  opp_ratings <- sapply(p$Opponents, function(oid) {
    if (!is.na(oid) && oid <= length(player_list) && !is.null(player_list[[oid]])) {
      player_list[[oid]]$PreRating
    } else {
      NA
    }
  })

  avg_rating <- round(mean(opp_ratings, na.rm = TRUE), 0)

  data.frame(
    Player_Name = p$Name,
    State = p$State,
    Total_Points = p$TotalPoints,
    Pre_Rating = p$PreRating,
    Avg_Opponent_Rating = avg_rating,
    stringsAsFactors = FALSE
  )
})
```

```r
# 6. Interactive Visualization
datatable(final_results,
          caption = "Chess Tournament Results: Player Performance Overview",
          options = list(pageLength = 10, searchHighlight = TRUE),
          filter = 'top')
```

Show 10 entries Search:

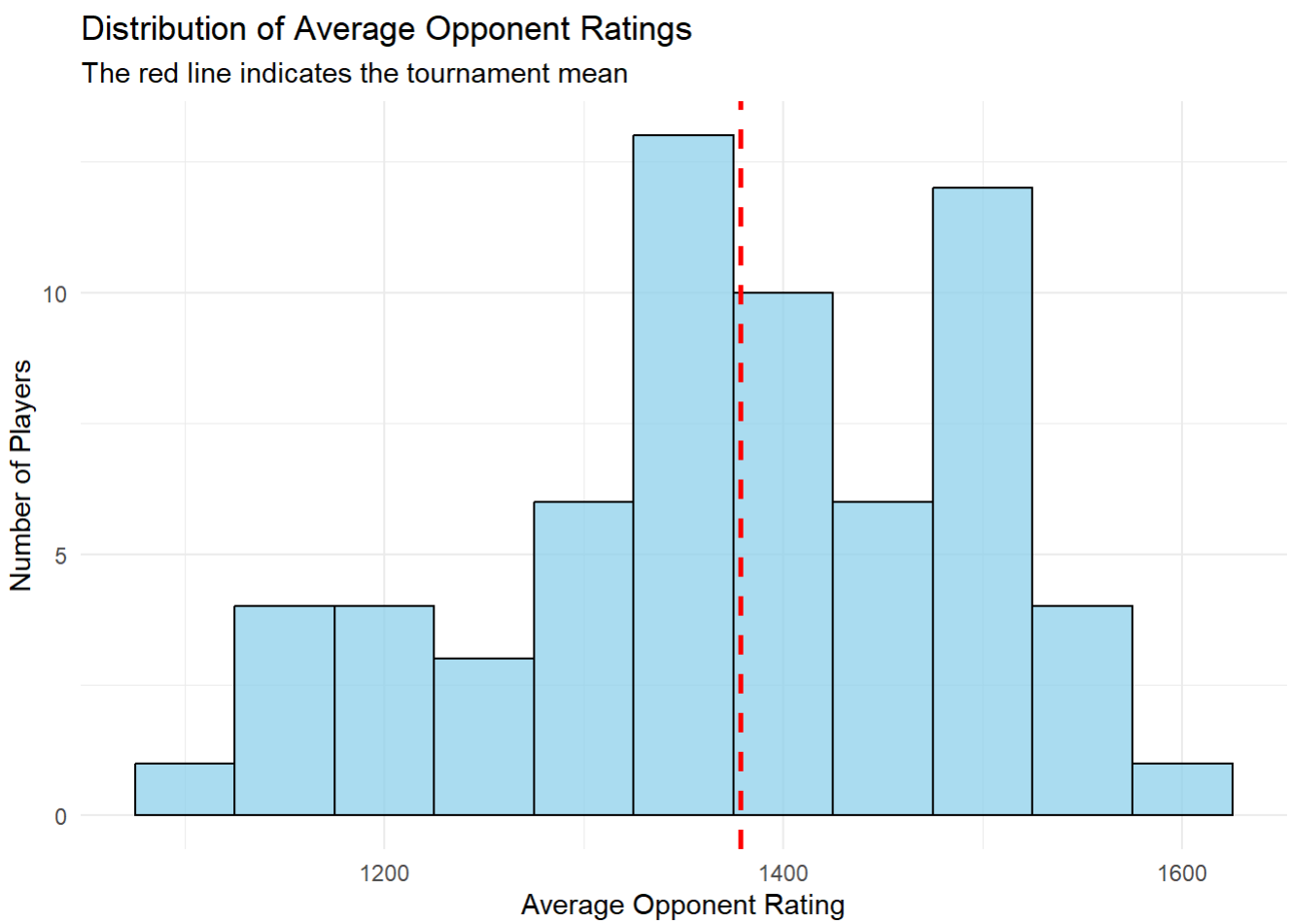
	Player_Name	State	Total_Points	Pre_Rating	Avg_Opponent_Rating
	All	All	All	All	All
1	GARY HUA	ON	6	1794	1605
2	DAKSHESH DARURI	MI	6	1553	1469
3	ADITYA BAJAJ	MI	6	1384	1564
4	PATRICK H SCHILLING	MI	5.5	1716	1574
5	HANSHI ZUO	MI	5.5	1655	1501
6	HANSEN SONG	OH	5	1686	1519
7	GARY DEE SWATHELL	MI	5	1649	1372
8	EZEKIEL HOUGHTON	MI	5	1641	1468
9	STEFANO LEE	ON	5	1411	1523
10	ANVIT RAO	MI	5	1365	1554

Chess Tournament Results: Player Performance Overview

Showing 1 to 10 of 64 entries Previous 1 2 3 4 5 6 7 Next

```r
# 7. Histogram
ggplot(final_results, aes(x = Avg_Opponent_Rating)) +
  geom_histogram(binwidth = 50, fill = "skyblue", color = "black", alpha = 0.7) +
  geom_vline(aes(xintercept = mean(Avg_Opponent_Rating, na.rm = TRUE)),
             color = "red", linetype = "dashed", size = 1) +
  labs(title = "Distribution of Average Opponent Ratings",
       subtitle = "The red line indicates the tournament mean",
       x = "Average Opponent Rating",
       y = "Number of Players") +
  theme_minimal()
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
ℹ Please use `linewidth` instead.
```




```r
# --- Расчет на основе FIDE Handbook, Section B.02 ---
# Источник: https://handbook.fide.com/chapter/B022017

# 1. Функция для расчета ожидаемого результата одной игры
calc_expected <- function(player_rtg, opp_rtg) {
  return(1 / (1 + 10^((opp_rtg - player_rtg) / 400)))
}

# 2. Добавляем расчеты в таблицу
final_results_elo <- map_df(player_list, function(p) {
  if (is.null(p)) return(NULL)

  # Получаем рейтинги реальных соперников
  opp_ratings <- sapply(p$Opponents, function(oid) {
    if (!is.na(oid) && oid <= length(player_list)) player_list[[oid]]$PreRating else NA
  })

  # Считаем сумму ожидаемых очков (только для сыгранных матчей)
  expected_scores <- sapply(opp_ratings, function(oa) {
    if (is.na(oa)) return(0)
    calc_expected(p$PreRating, oa)
  })

  total_expected <- round(sum(expected_scores, na.rm = TRUE), 2)
```

```r
  data.frame(
    Player_Name = p$Name,
    Actual_Points = p$TotalPoints,
    Expected_Points = total_expected,
    Diff = round(p$TotalPoints - total_expected, 2),
    stringsAsFactors = FALSE
  )
})

# 3. Пять лучших (Over-performed)
top_5_over <- final_results_elo %>% arrange(desc(Diff)) %>% head(5)
head(top_5_over)
```

```
               Player_Name Actual_Points Expected_Points Diff
1             ADITYA BAJAJ           6.0            1.95 4.05
2   ZACHARY JAMES HOUGHTON           4.5            1.37 3.13
3                ANVIT RAO           5.0            1.94 3.06
4 JACOB ALEXANDER LAVALLEY           3.0            0.04 2.96
5     AMIYATOSH PWNANANDAM           3.5            0.77 2.73
```

```r
# 4. Пять худших (Under-performed)
top_5_under <- final_results_elo %>% arrange(Diff) %>% head(5)
head(top_5_under)
```

```
         Player_Name Actual_Points Expected_Points  Diff
1   LOREN SCHWIEBERT           3.5            6.28 -2.78
2 GEORGE AVERY JONES           3.5            6.02 -2.52
3           JARED GE           3.0            5.01 -2.01
4       RISHI SHETTY           3.5            5.09 -1.59
5   JOSHUA DAVID LEE           3.5            4.96 -1.46
```